

Autosomal partly dominant inheritance

Sickle cell anemia

- Sickle cell anemia is an example of human disorders that is controlled by incompletely dominant allele.
- SCA occurs at a high rate among blacks (selective advantage)
- In persons with sickle cell trait (heterozygous HbA HbS) the hemoglobin contains both normal and sickle cell Beta polypeptide chains.

-They do not usually have any sickle-shaped cells unless they experience dehydration or mild oxygen deprivation; when subjected to extreme exercise such persons are more likely to die

-Persons with sickle cell disease (homozygous HbS HbS) have irregular or sickle shaped RBCs. The defect is caused by an abnormal hemoglobin that accumulates inside the cells.

Mutation in Beta globin gene (GAG→GUG in mRNA) resulted in replacement of Glutamic acid by Valine at 6th position of 146 amino acid sequence of Beta globin chain.

-Because sickle shaped cells are less flexible and are unable to pass along narrow capillary like disk shaped cells they clog the blood vessels and break down.

-This resulting in poor circulation due to vascular obstruction that produces localized hypoxemia and infarctions of various internal organs, including bone, kidneys and lungs. Premature destruction of the sickle erythrocytes produces anemia and jaundice

Two copies of the sickle-cell allele

All hemoglobin is the sickle-cell (abnormal) variety

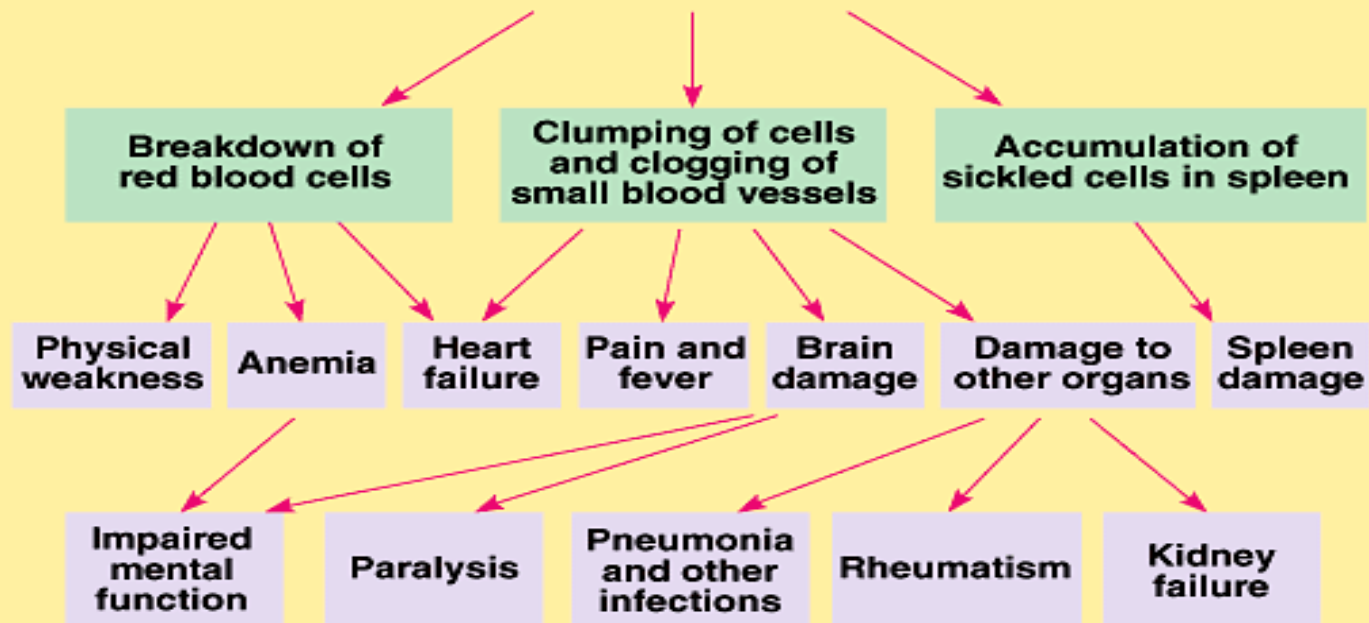
Abnormal hemoglobin crystallizes when oxygen content of blood is low, causing red blood cells to become sickle-shaped



Normal cells



Sickled cells



Extra nuclear inheritance (Mitochondria inheritance)

- Each human cell contains several hundred or more mitochondria dispersed through out the cytoplasm, they have their own DNA molecules, occurring in several copies per organelle.
- The mitochondrial DNA encodes two ribosomal RNA, 22t RNA and 13 polypeptides involved in oxidative phosphorylation to produce ATP.

- Mitochondrial enzymes are usually encoded by nuclear genes, but some of cytochrome oxidase enzymes are derived solely from mitochondria.
- Because it is located in the cytoplasm, mt DNA is inherited exclusively through the maternal line.
- Males cannot pass it to their offspring because sperm cells contain only a few mt DNA that do not enter the egg.

- Mutations involving the mt DNA are known to account for a small number of genetic diseases.
- One of the best known mitochondrial diseases is (LHON) Leber Hereditary Optic Neuropathy; this disease is characterized by rapid loss of vision as a result of optic nerve death.

-Vision loss typically begins in the 3rd decade of life and is irreversible.

-**MERRF**(ataxia,deafness, dementia,epileptic seizure and myopathy).

-**KSS**(paralysis of the eye muscles ,drooply eyelid, vision loss,deafness, and dementia).

MELAS(muscle weakness, pain, recurrent headache, loss of appetite and seizure).

- It has been suggested that mt DNA mutations which accumulative through out the life of the individual as a result of free radical formation, could contribute to the aging process

Multifactorial inheritance & common diseases (disorders):

- Multifactorial meaning that both genetic and environmental factors are collectively influence phenotype.
- A child of tall parents is more likely to be taller than is a child of short parents, but poor nutrition (an environmental factor) may compromise the growth of either child.

- Most quantitative phenotype variations among normal individuals are due to multifactorial traits.
- Examples include the range of skin pigmentation, weight, height and intelligence.

- Many human diseases are also inherited as multifactorial traits.
- The long list of M.D, includes cleft lip and/or palate, clubfoot, congenital dislocation of the hips, hypertension, heart disease, arteriosclerosis, diabetes mellitus, allergies, cancers, obesity and some forms of mental illness, such as schizophrenia and manic \ depressive psychosis.

- Many genes affect our cardiovascular health, making some of us more prone than others to heart attacks and strokes.
- But a healthful diet, exercise, lack of smoking and an ability to put stressful situations in perspective all reduce our risk of heart disease.